

NETWORK ANALYSIS

- DIRECTED GRAPH
- UNDIRECTED GRAPH
- ADJACENCY MATRIX
- EDGE LIST
- ADJACENCY LIST
- PLANAR GRAPH
- GEODESIC GRAPH
- GRAPH ECCENTRICITY
- GRAPH DIAMETER
- EULERIAN PATH
- HAMILTONIAN PATH

- DEGREE CENTRALITY
- DEGREE CENTRALITY UNDIRECTED GRAPH
- DEGREE DISTRIBUTION
- DENSITY
- BETWEENNESS
- CLOSENESS
- CONNECTED COMPONENT
- STRONGLY CONNECTED COMPONENT
- WEAKLY CONNECTED COMPONENT
- GIGANT COMPONENT
- STRONG AND WEAK TIES
- CLUSTERING (GLOBAL AND LOCAL)
- STRUCTURAL HOLES

- RANDOM GRAPHS
 - ERDOS-RENYI
 - WATTS-STROGATZ MODE
 - BARABASI-ALBERT MODEL
 - BARABASI-BIANCONI MODEL

- SCALE-FREE PROPERTIES
- RANDOM WALK MODEL
- THE WEB GRAPH
- SEARCH ENGINES
 - WEB CRAWLERS
 - BFS ARLGRITHM
 - RANK
 - TF-IDF
 - COSINE SIMILARITY

- MATH BACKGROUND FOR PageRank - MARKOV PROCESS
- GOOGLE PageRank
- TOPIC-SENSITIVE PageRank
- WEB SPAM
- CO-CITATION & BIBLIOGRAPHIC COUPLING & CO-REFERENCE
- HUBS AND AUTHORITIES - HITS

- PATTERNS OF RELATIONS
 - STRUCTURAL EQUIVALENCE
 - AUTOMORPHIC EQUIVALENCE
 - SIMILARITY MEASURES
 - JACCARD SIMILARITY
 - COSINE SIMILARITY
 - PEARSON CORRELATION COEFFICIENT

ASSORTATIVITY

- HOMOPHILY
- ASSORTATIVE AND DISASSORTATIVE MIXING
- ASSORTATIVE AND DISASSORTATIVE NETWORKS

COMMUNITIES IN NETWORKS

- ZACHARY KARATE CLUB
- GRAPH BISECTION
- RECURSIVE BISECTION
- MULTI-LEVEL GRAPH PARTITIONING
- MUTUALITY OF TIES: MAXIMUM CLIQUES
- FREQUENCY OF TIES: STRONG AND WEAK COMMUNITIES
- FREQUENCY OF TIES: K-CORES
- CORE DECOMPOSITION
- CLOSENESS OR REACHABILITY: K-DISTANCE CLIQUES
- COMMUNITY DETECTION
- LABEL PROPAGATION
- THE GIRVAN-NEWMAN ALGORITHM
- MODULARITY VS ASSORTATIVITY
- LOUVAIN METHOD
- LEIDEN ALGORITHM

INFOMAP

- OVERLAPPING COMMUNITIES
- SPEAKER-LISTENER LABEL PROPAGATION ALGORITHM

NETWORK DIFFUSION: EPIDEMIC SPREADING

- SIS EPIDEMIC MODEL
- SIR EPIDEMIC MODEL
- SIRS EPIDEMIC MODEL
- SIRS EPIDEMIC MODEL + SMALL WORLD

DYNAMICS: SOCIAL CONTAGION

- SIMPLE CONTAGION
- COMPLEX CONTAGION
- THRESHOLD MODEL

NETWORK ROBUSTNESS

- PERCOLATION THEORY
- INVERSE PERCOLATION THEORY

ROBUSTNESS OF SCALE-FREE NETWORKS

- MOLLOY-REED CRITERION
- BARABASI ET AL. EXPERIMENT

CASCADING FAILURES

- FAILURE PROPAGATION MODEL
- BRANCHING MODEL
- MOTTER AND LAI MODEL